

Database System Engineering and Distributed Backend Development

WEEK-2

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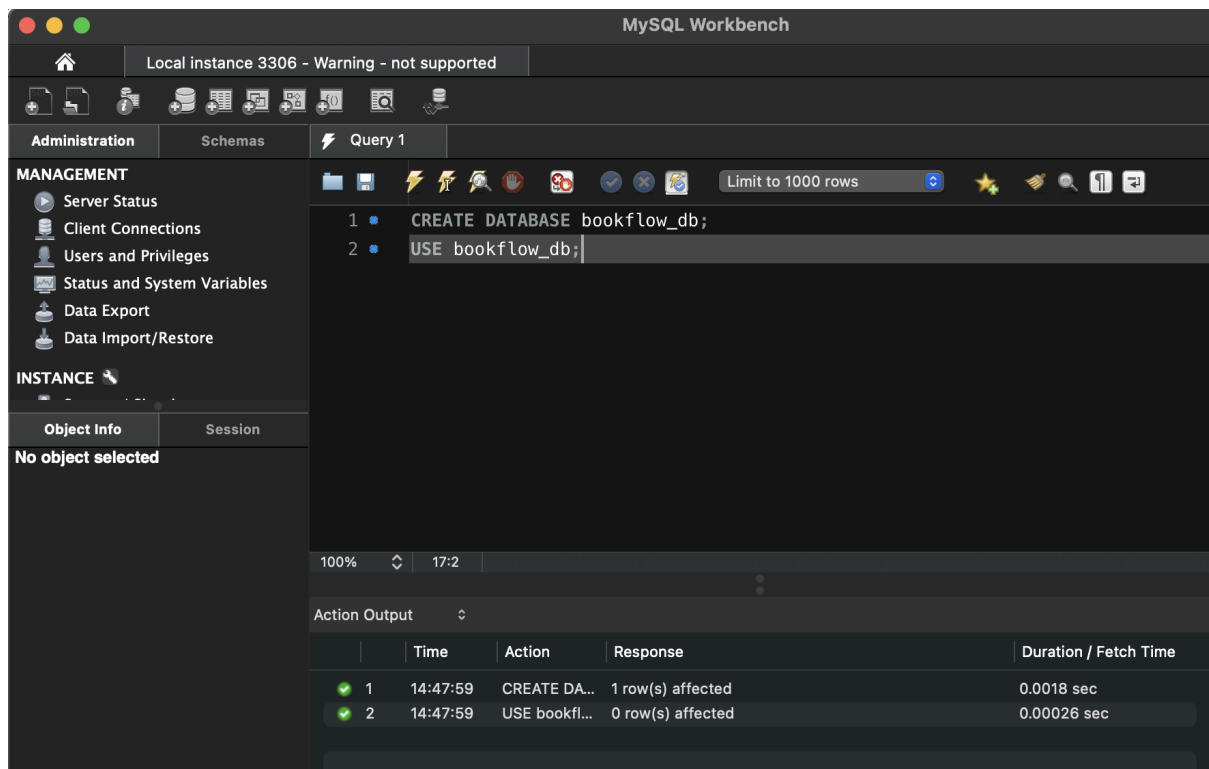
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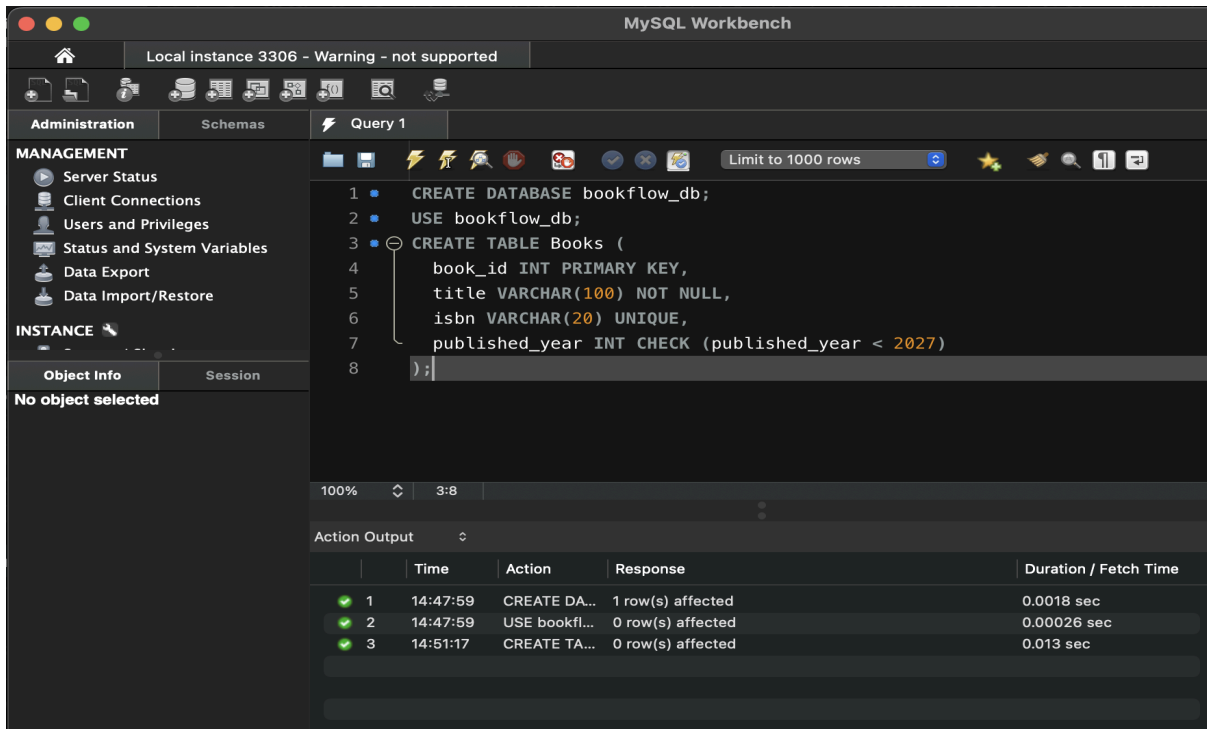
Step 1 — Create Database

```
CREATE DATABASE bookflow_db;  
USE bookflow_db;
```



Step 2 — Create Books Table (DDL + Constraints)

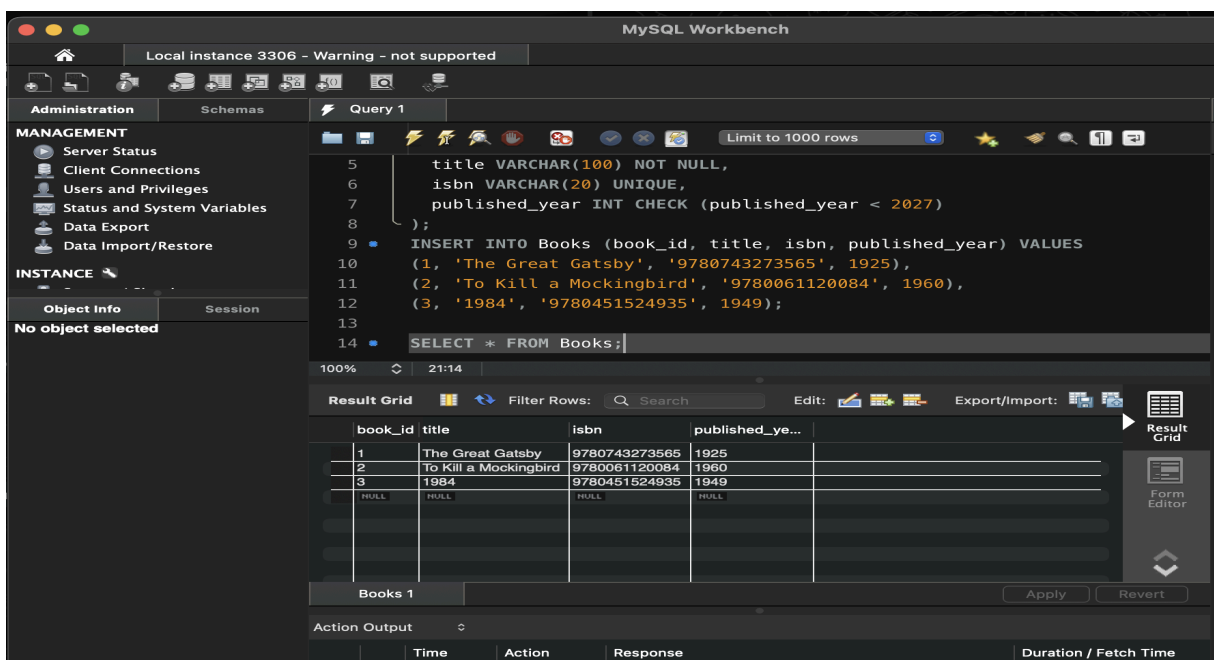
```
CREATE TABLE Books (  
  book_id INT PRIMARY KEY,  
  title VARCHAR(100) NOT NULL,  
  isbn VARCHAR(20) UNIQUE,  
  published_year INT CHECK (published_year < 2027)  
);
```



Step 3 & 4 — Insert Books and Display

```
INSERT INTO Books (book_id, title, isbn, published_year) VALUES
(1, 'The Great Gatsby', '9780743273565', 1925),
(2, 'To Kill a Mockingbird', '9780061120084', 1960),
(3, '1984', '9780451524935', 1949);

SELECT * FROM Books;
```



Step 5, 6 & 7 — Members Table: Create, Insert, Display

```
CREATE TABLE Members (  
    member_id INT PRIMARY KEY,  
    full_name VARCHAR(100),  
    email VARCHAR(100) UNIQUE  
);  
  
INSERT INTO Members (member_id, full_name, email) VALUES  
(101, 'John Smith', 'john.smith@email.com'),  
(102, 'Emma Wilson', 'emma.wilson@email.com'),  
(103, 'Michael Brown', 'michael.brown@email.com');  
  
SELECT * FROM Members;
```

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. Below this is the 'INSTANCE' section with Startup / Shutdown and Server Logs. The 'Object Info' section shows 'No object selected'. The main query editor displays the following SQL code:

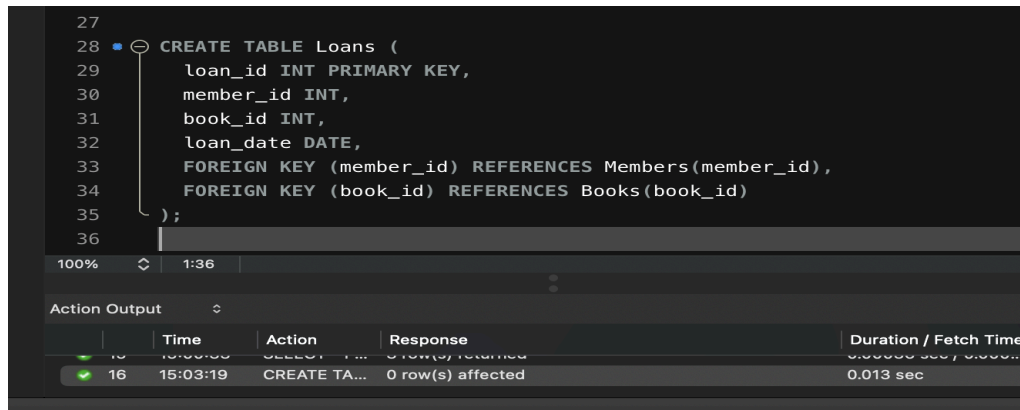
```
14 SELECT * FROM Books;  
15  
16 CREATE TABLE Members (  
17     member_id INT PRIMARY KEY,  
18     full_name VARCHAR(100),  
19     email VARCHAR(100) UNIQUE  
20 );  
21  
22 INSERT INTO Members (member_id, full_name, email) VALUES  
23     (101, 'Hansika', 'gunapatihansikareddy@klh.edu.in'),  
24     (102, 'Hasini', 'hasinireddy@klh.edu.in'),  
25     (103, 'Bindhu', 'bindhumadhavi@klh.edu.in');  
26  
27 SELECT * FROM Members;
```

The 'Result Grid' at the bottom shows the output of the last query:

member_id	full_name	email
101	Hansika	gunapatihansikareddy@klh.edu.in
102	Hasini	hasinireddy@klh.edu.in
103	Bindhu	bindhumadhavi@klh.edu.in

Step 8 — Create Loans Table (FOREIGN KEYS)

```
CREATE TABLE Loans (  
  loan_id INT PRIMARY KEY,  
  member_id INT,  
  book_id INT,  
  loan_date DATE,  
  FOREIGN KEY (member_id) REFERENCES Members(member_id),  
  FOREIGN KEY (book_id) REFERENCES Books(book_id)  
);
```

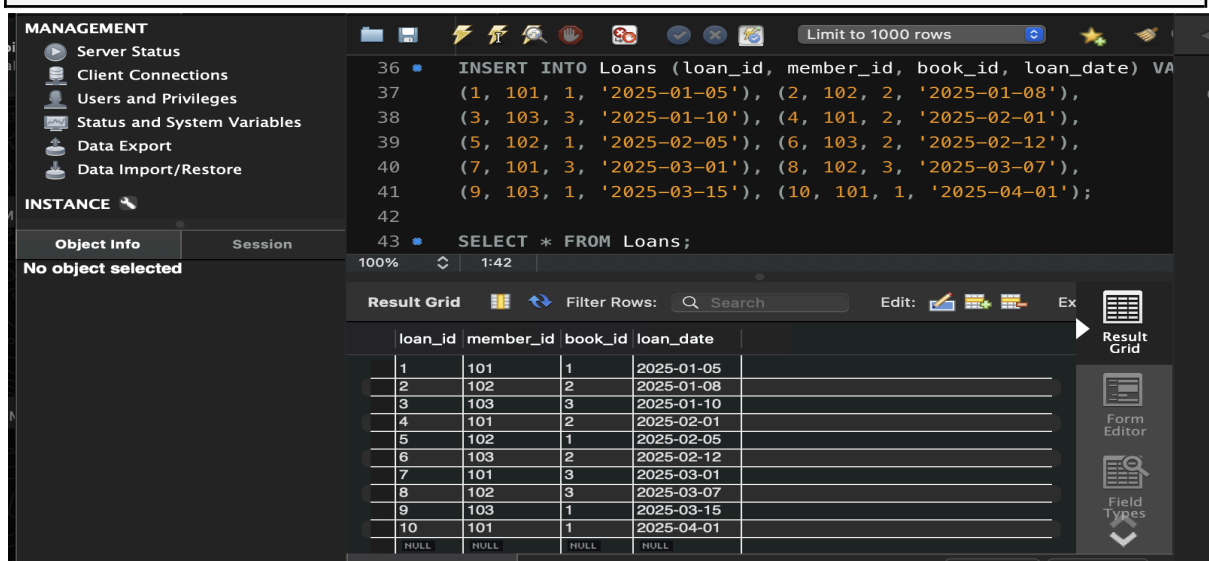


The screenshot shows a SQL IDE with a dark theme. The SQL editor contains the same CREATE TABLE Loans statement as in Step 8. Below the editor, the 'Action Output' pane shows a table with columns: Time, Action, Response, and Duration / Fetch Time. The first row shows the execution of the CREATE TABLE statement at 15:03:19, with the response '0 row(s) affected' and a duration of 0.013 sec.

Time	Action	Response	Duration / Fetch Time
15:03:19	CREATE TA...	0 row(s) affected	0.013 sec

Step 9 & 10 — Insert 10 Loans and Display

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES  
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),  
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),  
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),  
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),  
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');  
  
SELECT * FROM Loans;
```



The screenshot shows a database management tool interface. The left sidebar has a 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. Below this is an 'INSTANCE' section with 'Object Info' and 'Session' tabs. The main area shows the SQL editor with the INSERT and SELECT statements. Below the editor, the 'Result Grid' pane displays the results of the SELECT statement. The grid has columns for loan_id, member_id, book_id, and loan_date, and it shows 10 rows of data.

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

Step 11 — JOIN Query (Who Borrowed What)

```
SELECT m.full_name AS Member_Name, b.title AS Book_Title
FROM Loans l
INNER JOIN Members m ON l.member_id = m.member_id
INNER JOIN Books b ON l.book_id = b.book_id;
```

The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying a SQL query that joins the 'Loans', 'Members', and 'Books' tables. The query is as follows:

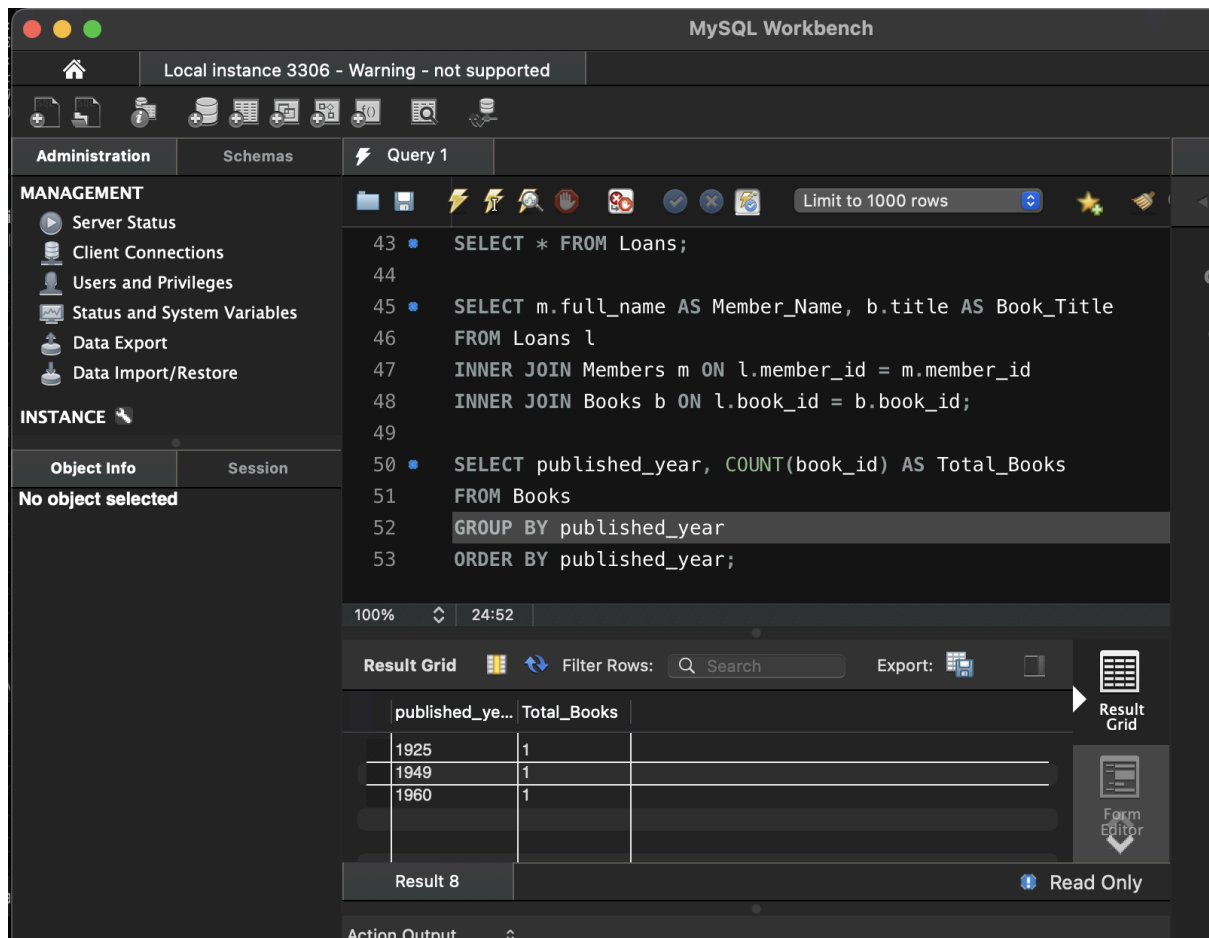
```
SELECT * FROM Loans;
SELECT m.full_name AS Member_Name, b.title AS Book_Title
FROM Loans l
INNER JOIN Members m ON l.member_id = m.member_id
INNER JOIN Books b ON l.book_id = b.book_id;
```

The 'Result Grid' is visible at the bottom, showing the results of the query. The results are as follows:

Member_Name	Book_Title
Bindhu	The Great Gatsby
Hasini	The Great Gatsby
Hansika	The Great Gatsby
Hansika	The Great Gatsby
Bindhu	To Kill a Mockingbird
Hasini	To Kill a Mockingbird
Hansika	To Kill a Mockingbird
Bindhu	1984
Hasini	1984
Hansika	1984

Step 12 — GROUP BY Query (Books per Year)

```
SELECT published_year, COUNT(book_id) AS Total_Books
FROM Books
GROUP BY published_year
ORDER BY published_year;
```



Step 13–16 — Donation via TRANSACTION (Atomicity)

```

CREATE TABLE Donation_History (
    donation_id INT PRIMARY KEY,
    book_id INT,
    donor_name VARCHAR(100),
    donation_date DATE,
    FOREIGN KEY (book_id) REFERENCES Books(book_id)
);

START TRANSACTION;

INSERT INTO Books (book_id, title, isbn, published_year)
VALUES (4, 'Animal Farm', '9780451526342', 1945);

INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
VALUES (1, 4, 'Raj Kumar', CURDATE());

COMMIT;

```

MySQL Workbench

Local instance 3306 - Warning - not supported

Administration Schemas Query 1

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

Object Info Session

No object selected

```

52 GROUP BY published_year
53 ORDER BY published_year;
54
55 CREATE TABLE Donation_History (
56     donation_id INT PRIMARY KEY,
57     book_id INT,
58     donor_name VARCHAR(100),
59     donation_date DATE,
60     FOREIGN KEY (book_id) REFERENCES Books(book_id)
61 );
62
63 START TRANSACTION;
64
65 INSERT INTO Books (book_id, title, isbn, published_year)
66 VALUES (4, 'Animal Farm', '9780451526342', 1945);
67
68 INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
69 VALUES (1, 4, 'Raj Kumar', CURDATE());
70
71 COMMIT;

```

100% 8:71

Action Output

	Time	Action	Duration / Fetch Time
24	15:11:11	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', '9780451526342', 1945);	0.00017 sec
25	15:11:11	COMMIT	0.00049 sec

Query Completed

Step 17 & 18 — Index on ISBN + Fast Search

```

CREATE INDEX idx_books_isbn ON Books(isbn);

SELECT * FROM Books WHERE isbn = '9780451524935';

```

Object Info Session

No object selected

```

72
73 CREATE INDEX idx_books_isbn ON Books(isbn);
74
75 SELECT * FROM Books WHERE isbn = '9780451524935';

```

100% 50:75

Result Grid

book_id	title	isbn	published_year
3	1984	9780451524935	1949

Books 9

Apply Revert